Peripheral Device: ~~up এর আগের exter~~

Peripherals are any device that can be attached to a CPU for example hard disk, mouse, printer. Peripherals are slow. They hold up the CPU from carrying out its tasks.

Interface:

The interface is the combination of hardware and software needed to link the CPU to the peripherals and to enable them to communicate with CPU despite all their deiffering characteristics.

Computer peripherals have different characteristic:

→ Have different data transfer rate

→ Use a wide variety of codes and control signals.

→ Transmit data in serial or in parallel form.

→ Even work at higher voltage than CPU

→ All operate at much slower speeds than CPU.

* The main functions of an interface :

→ Buffering

→ converting data to and from serial and parallal form

→ converting dat to and from analogue and digital forms.

→ voltage conversion

→ Protocol

→ Handling of status signal.

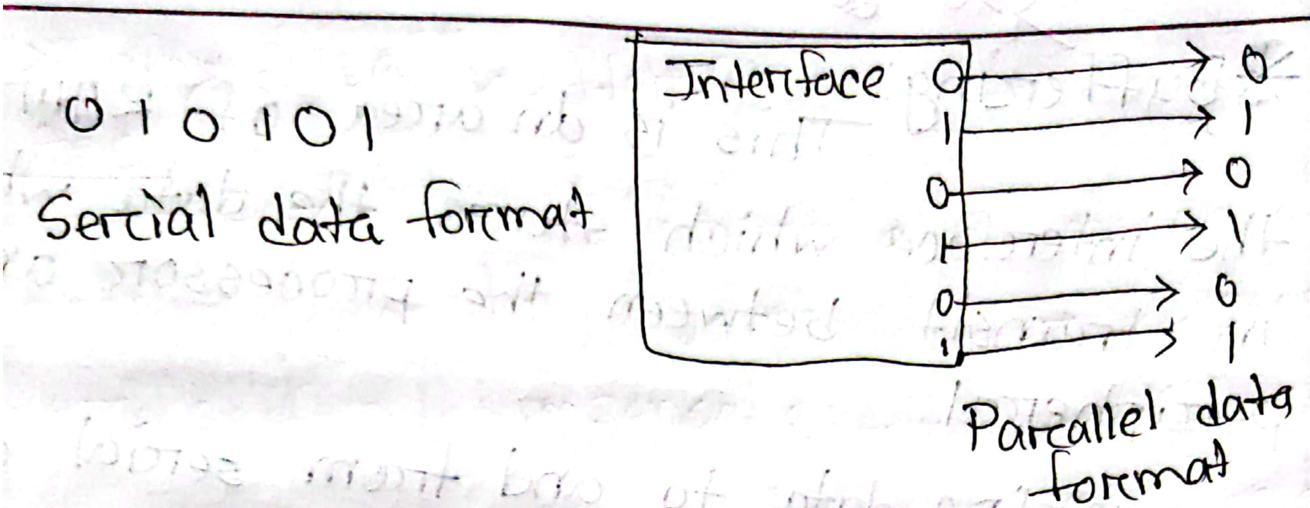
temporary data store করে

*Buffering: This is an area of RAM within the interface which stores the data while in transit between the processor and the peripheral

*converting data to and from serial and parallel forms: Data transmission is the passing of data from one device to another.

serial data transmission: is when data is transmitted along a communication channel one bit after, in sequence. Very slow but efficient over long distance.

⇒ Parallel data transmission: transmit several bits of data simultaneously across a series of parallel channels, often transmitting 16 or 32 bits at a time. The buses internal to the processor are parallel channels.



⇒ Converting data to and from analogue and digital forms:

→Analogue signal: Analogue signals that are sent in from peripherals to the digital form that the CPU can handle.

Digital signal: Computers can only work with digital signals, which have only two values - on or off.

ADC → Analogue to Digital converter

⇒ DAC: A computer is connected to a peripheral by an interface. This interface has to be able to change the digital signals from the computer to an analogue signal that the other device can understand. This is called Digital to Analogue Converter.

* Protocol Conversion: A protocol is a standard that enables connection, communication and data transfer between computers or between a computer system and peripheral. Protocol conversion means ensuring that protocols used by the peripheral can be understood by the computer it is attached to.

* Handling of Status signal: The purpose of status information is to show whether or not a peripheral device is ready to communicate.

* Why use buffer?

Ans:

→ Peripherals operate at much slower speeds than CPU. Using buffer helps computer system compensate for the differences in operating speeds between CPU and its peripheral.

→ When transferring data out to a peripheral such as a printer, the processor can transfer it to buffer and buffer will send it to the printer at a speed that printer can cope with.

→ The use of buffer reduces the frequency with which the CPU is interrupted to deal with input.

CamScanner

*Spoolers: Spooling is another technique used in the transfer of data to a slow peripherals. For example in printer data is transferred to storage, often a hard disk. Then when the processor is idle it will transfer the data to the printer at a acceptable speed. This is also called background printing.

* Buffer vs	Spooler.
1. If CPU is very busy and doesn't get much idle time buffer is best to use.	1. If CPU is very busy spooling can be very slow process.
2. Buffer is limited by the amount of RAM	2. Spooler uses backing storage which has large capacity.

Use both ~~use both~~ for optimum efficiency.

Buffering	Spooling
3. It needs greater resource management.	3. It needs less resource management
4. More efficient	4. Less efficient
5. It overlaps the I/O of one ~~book~~ task with the computation of the same task.	5. It overlaps the I/O of one task with the computation of another task.

*Wifi – Wireless Fidelity Alliance.

* Flash card: Flash card are used mainly for data storage in cameras, although they can hold any type of program or data file.

* Microprocessor Data Type:

→ Bit

→ Byte

→ Word

→ Unsigned and signed Binary Integers.

→ BCD

→ ASCII

→ Floating Point Numbers.

* Mapping : Signed ⇔ Unsigned

4 bit এর জন্য

Bits	Signed	Unsigned
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7
1000	-8	8
1001	-7	9
1010	-6	10
1011	-5	11
1100	-4	12
1101	-3	13
1110	-2	14
1111	-1	15

Floating Point Number

Date: ..../......../...........

* The conversion Procedure:

→ Convert the absolute value of the number to binary, perhaps with a fractional part after binary point.

→ Append $\times 2^0$ to the end of the binary number which does not change its value.

→ Normalize the number. Move binary point so that it is one bit from the left. Adjust exponent of two so that the value does not change.

→ Place mantissa into the mantissa field of the number. Omit the leading one, and fill with zeros on the right.

→ Add the bias to the exponent of two and place it in the exponent field. The bias is $2^{(k-1)}-1$, where k is the number of bits in the exponent field. For eight bit format k=3, so bias $2^{(3-1)}-1=3$. For IEEE 32-bit, k=8, so $2^{(8-1)}-1=127$.

→ Set the sign bit, 1 for negative, 0 for positive, according to the sign of the original number.

8bit format এর জন্য exponent always 3bit
〃 〃 〃 〃 sign 1 bit
〃 〃 〃 〃 mantissa 4bit

*The range of sign magnitude form is from $-(2^{(n-1)}-1)$ to $(2^{(n-1)}-1)$

*Floating Point Number: It contains three components :– sign, exponent and mantissa

For the decimal value -2.5×10^{-2},
sign is negative,
exponent is -2
mantissa is 2.5

* convert 2.625 to our $\boxed{8 \text{ bit}}$ floating point format.

এখানে first এ বলা এই দেওয়া হবে কত bit format বলবে

Ans:
1. The integral part is $2_{10} = 10_2$

For fractional part:

$$0.625 \times 2 = 1.25 \quad 1$$
$$0.25 \times 2 = 0.5 \quad 0$$
$$0.5 \times 2 = 1.0 \quad 1$$

so, $(0.625)_{10} = (0.101)_2$ and $(2.625)_{10} = (10.101)_2$

Decimal এর জন্য exponent power = 10
Binary " " " " = 2

2. Add an exponent part: $(10.101)_2 = 10.101 \times 2^0$

positive নাম্বার যে কয়ঘর বামে নিব exponent এ তত যোগ হবে

3. Normalize: $10.101 \times 2^0 = 1.0101_2 \times 2^1$

দশমিক সংখ্যাকে এমনভাবে change করতে হবে যাতে দশমিকের বামে ১টি সংখ্যা থাকে।

4. Mantissa : 0101

দশমিকের পরের value mantissa হিসেবে কাজ করবে

5. Exponent : $\underbrace{1}_{2 \text{ এর power } 1} + 3 = 4 = 100_2$

8 bit এর জন্য bias value 3
32 " " " " 127

6. Sign bit is 0

The result is 0 100 0101

Represented as hex, that is 45_{16}

*. Convert -4.75 to our 8-bit floating point format.

Ans: 1. The integral part is $4_{10} = 100_2$

For fractional part:

$0.75 \times 2 = 1.25$ | 1

$0.25 \times 2 = 0.5$ | 1

$0.5 \times 2 = 1.0$ | 1

So, $0.75_{10} = 0.101_2$ and $4.75_{10} = 1000.101_2$

2. Add an exponent part: 1000.101×2^0

3. Normalize: $1000.101 \times 2^0 = 1.000101 \times 2^3$

4. Mantissa: 000101

5. Exponent: $3+3=6=0110$

6. Sign bit is: 1

The result is:

1 110 0101

*Register

⇒ In unsigned arithmetic we need to watch carry flag.

⇒ In signed arithmetic we need to watch overflow flag.

*The carry flag is set if:

→ the addition of two numbers causes a carry out of the most significant (leftmost) bits added. For example:

1111 + 0001 = 0000 ⇒ carry flag is turned on.

→ the ~~s~~ subtraction of two numbers requires a borrow into the most significant (leftmost) bits subtracted. For example:

0000 − 0001 = 1111 ⇒ carry flag turned on.

*The overflow Flag is set if:

→ the sum of two numbers with sign bit off yield a result number with the sign bit on. For example:

0100 + 0100 = 1000 ⇒ overflow flage turned on.

→ the sum of two numbers with the sign bit on yields a result number with sign bit off. For example:

1000 + 1000 = 0000 ⇒ overflow flag turned on

*যখন দুটি positive integer যোগ করা হয় এবং leftmost bit 1 আসে, তখন সেটা overflow flag turned on হয় কারণ 1 যেহেতু negative বুঝায় এবং দুটি positive যোগ করে কখনও negative আসে না তাই processor overflow flag on করে।

যেমন = 0100 + 0100 = $\boxed{1}$000
↳ overflow on.

Mixed sign addition never turns on overflow flag. always 2টা number positive বা negative signed হলে হবে।

For example:

assume binary addition 0111+0010 = 1001

→ if we are doing unsigned mathematics

0111 + 0010 = 1001 → overflow turned on

7 + 2 = 9

→ if we ~~are~~ do signed mathematics

0111 + 0010 = 1001 → overflow on

7 + 2 = −7 → error detected

এজন্য processor এই math unsignedএ

যোগ করে।

* A 1K memory device has 10 address Pins (A0−A9)

* A 16K×1 is a memory device containing 16K 1 bit memory location.

* 64K×4 memory device is listed as a 256K device.

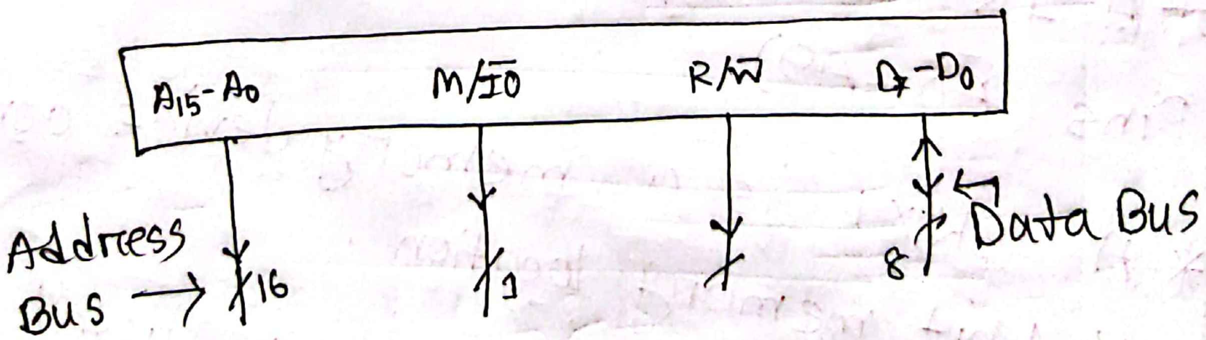
bit হারে possible.

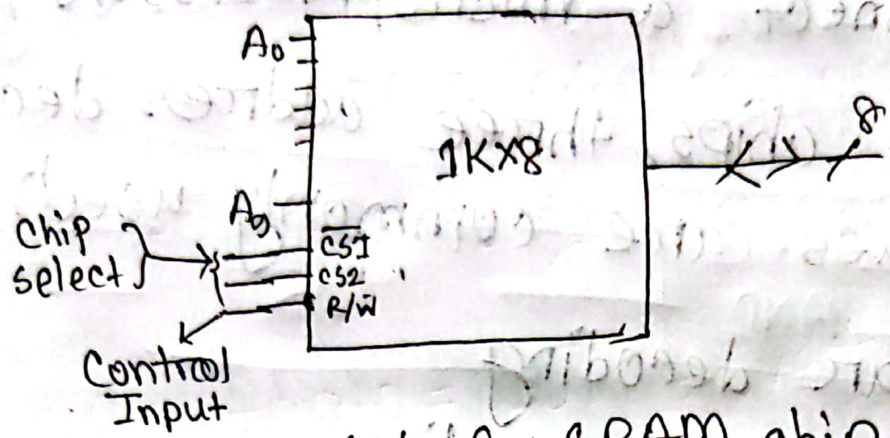
*Address decoding: It refers to the way a computer system decodes the address on address bus to select memory location in one or more memory or peripheral device

*Main memory array design:

M/$\overline{IO}$ → LOW: if microprocessor executes an I/O instruction.

M/$\overline{IO}$ → HIGH: if microprocessor executes a memory instruction





1K x 8 SRAM chip

1000 — 8bit

1000 আকআছে

1000 আকাকে একটা করে address দিব. 10 bit করে নিলে 1000 করা possible

Table

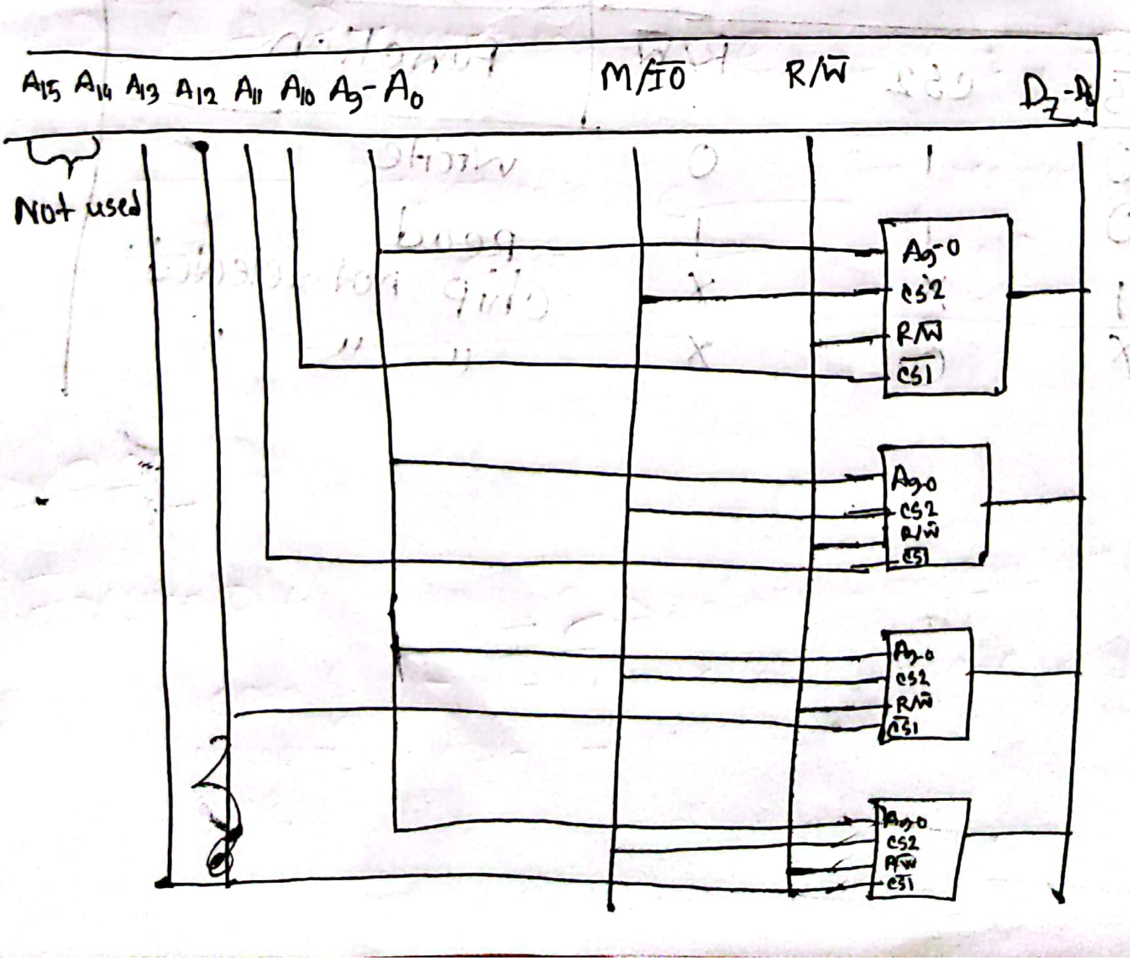
$\overline{CS1}$	CS2	$R/\overline{W}$	Function
0	1	0	write
0	1	1	Read
1	X	X	chip not selected
X	0	X	" " "

0 থাকতে হবে, না থাকলে don't care condition এ চলে যাবে

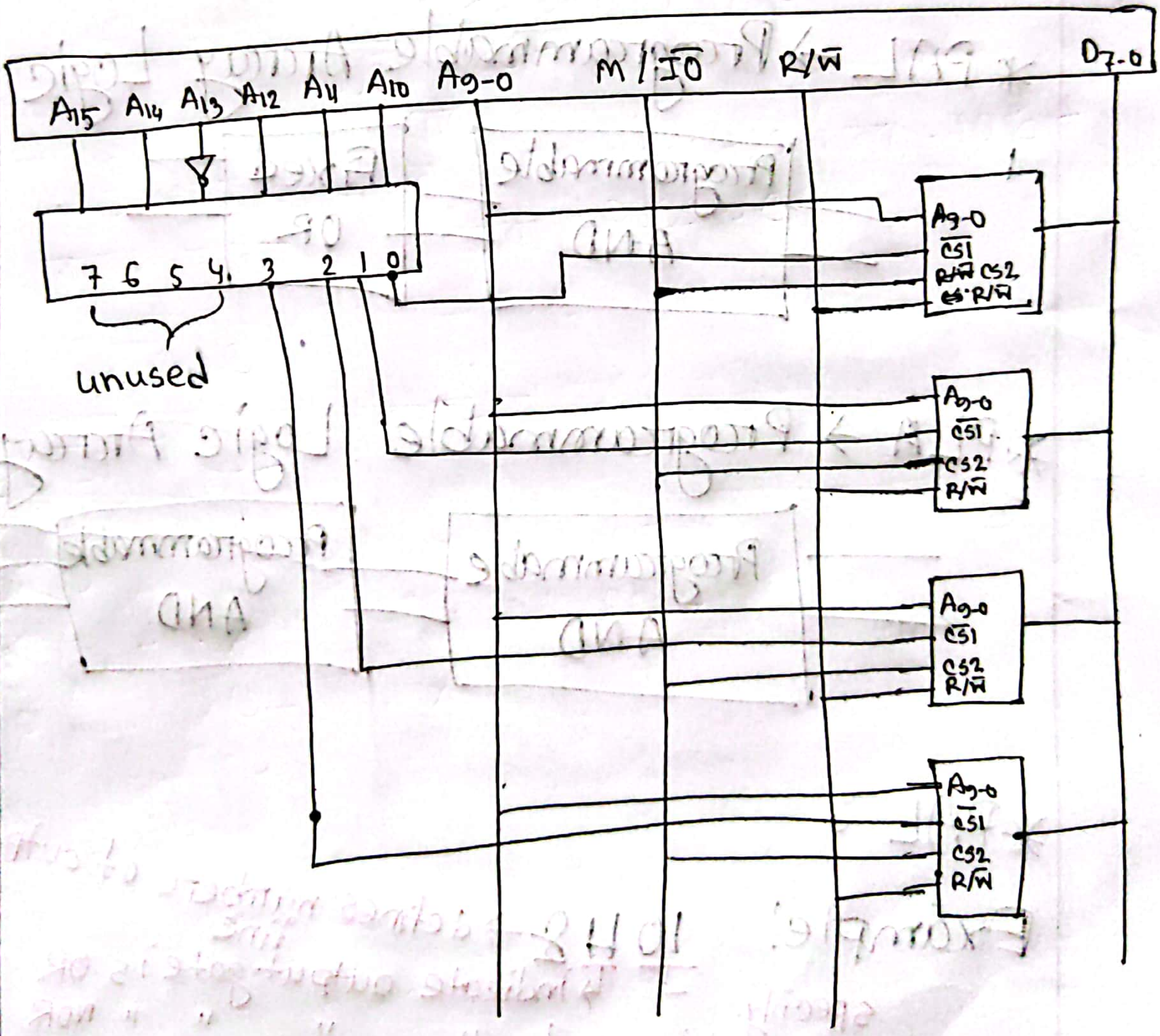
*To connect a microprocessor to ROM/RA chips, three address decoding techniques are commonly used:

→ Linear decoding
→ Full/Partial decoding
→ memory decoding using PALs.

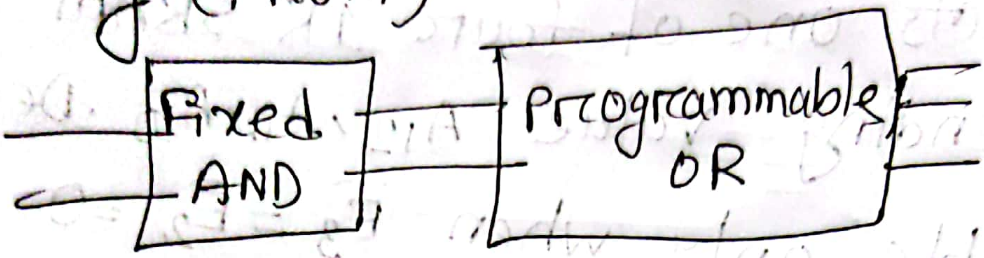
*Linear decoding:
Suppose we have 4K SRAM array comprised of four 1K.



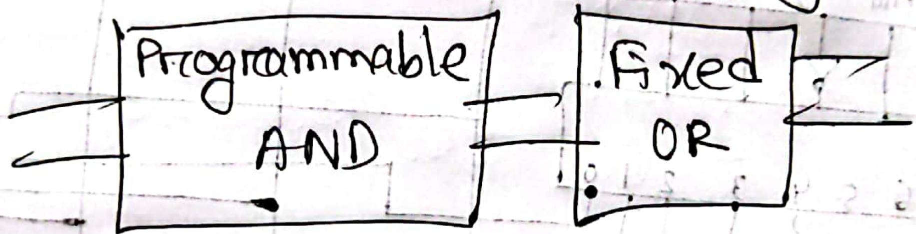
*Full decoding: using 3x8 decoder output selects one of four 1K SRAM chips depending value A_{12}, A_{11}, A_{10}. Decoder output enable only when $\overline{E_3} = \overline{E_2} = 0$ and $E_1 = 1$.



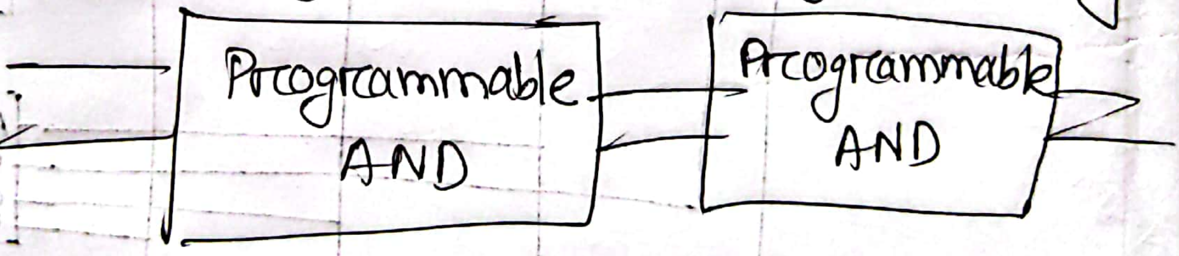
* PROM → Programmable Read Only memory (PROM)



* PAL → Programmable Array Logic



* PLA → Programmable Logic Array



* PAL :

Example: 10 H 8 → defines number of output line

10 → specify number of input line

H → indicate output gate is OR

L " " " " NOR

64K = 2^{16}

8085 is 64KB addressable. ~~16 bit এর~~ address generate করতে পারে. প্রতিটি 64K address যে ঘর নির্দেশ করবে তাতে 8 bit থাকতে পারবে. $A_0 - 15$

Example-01: A processor is having 16 bit address line and it is byte addressable. Interface 4096 × 8 RAM with the processor.

1. What is the memory capacity of processor
2. How many address lines are required for a RAM
3. How many RAM chips are required
4. What will be the size of address decoder

indicate করে কত bit লাগবে address bus এ. Normally A_0-15 লাগে

Address এ যে bit লাগে না তা CS অর্থাৎ chip select এ লাগে.

address হবে 16 bit-এর

4K = 4×1024 = 4096 = 2^{12}

~~Ans:~~

[* যদি 8085 → 16 bit address line উল্লেখ করা থাকে, তাহলে multiplexing, demultiplexing latch এর ব্যবহার উল্লেখ করে দেখাতে হবে এবং draw করতে হবে. 8086 এর 20 bit address line]

[** সব ক্ষেত্রে interface draw করে দিতে হবে]

Ans) byte addressable মানে word size 1 byte = 8 bit. অর্থাৎ, প্রতি তাকে 8bit করে hold করতে পারবে।

1. প্রশ্নে বলা আছে 16 bit address line

তাই memory capacity ~~= 64KB~~ $= 2^{16}$

$~~= 2^{16}~~$ $= 64 KB$

2. Here ~~IS~~ 4096 × 8 RAM.

So address line = 4096

$= 2^{\boxed{12}}$

= 12 address line

3. Processor এ total কয়টা RAM chip লাগবে

we know total memory capacity

$= 2^{16}$ Byte

~~= 6~~

memory capacity of a RAM $= 2^{12}$

$\therefore$ RAM chip needed $= \frac{2^{16}}{2^{12}}$

$= 2^{4}$

$= 16$ RAM

$= 4K$ RAM chip

4. The size of address decoder $=$
4 to 16 decoder

Interface এর figure দিতে হবে

Example - 2 : Interface two chips of 4K RAM and 1 chip of 2K ROM starting address of RAM is 8000H.

Ans:

এখানে address দেয়া 8000H. এখানে 4 টা digit আছে তাই CPU এর $2^4 = 16$ bit address generate করতে পারবে।

4K RAM

$= 4 \times 1024$

$= 4096$

$= 2^{12}$

অর্থাৎ, A_{0-11} পর্যন্ত address দরকার নির্দিষ্ট RAM টাকে address করার জন্য।

2K ROM

$= 2 \times 1024$

$= 2048$

$= 2^{11}$

$\therefore$ 11 address লাগবে ROM কে address করার জন্য.

যেহেতু 4K RAM তাই 12 bit নিয়ে কাজ করছে

Chip selection

16

	A_{15}	A_{14}	A_{13}	A_{12}	A_{11}	A_{10}	A_9	A_8	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0
1st RAM First add 8000	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8FFF Last add	1	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1
2nd RAM 1st 9000	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
Last 9FFF	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1
ROM 1st A000	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
1st A800	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0
last A7FF AFFF	1	0	1	0	0	1	1	1	1	1	1	1	1	1	1	1

Chip selection

ROM এর জন্য 11 টা bit নিয়ে কাজ করবে

১ম RAM এর Last address এর সাথে 1 যোগ করলে ২য় RAM এর first add আসবে

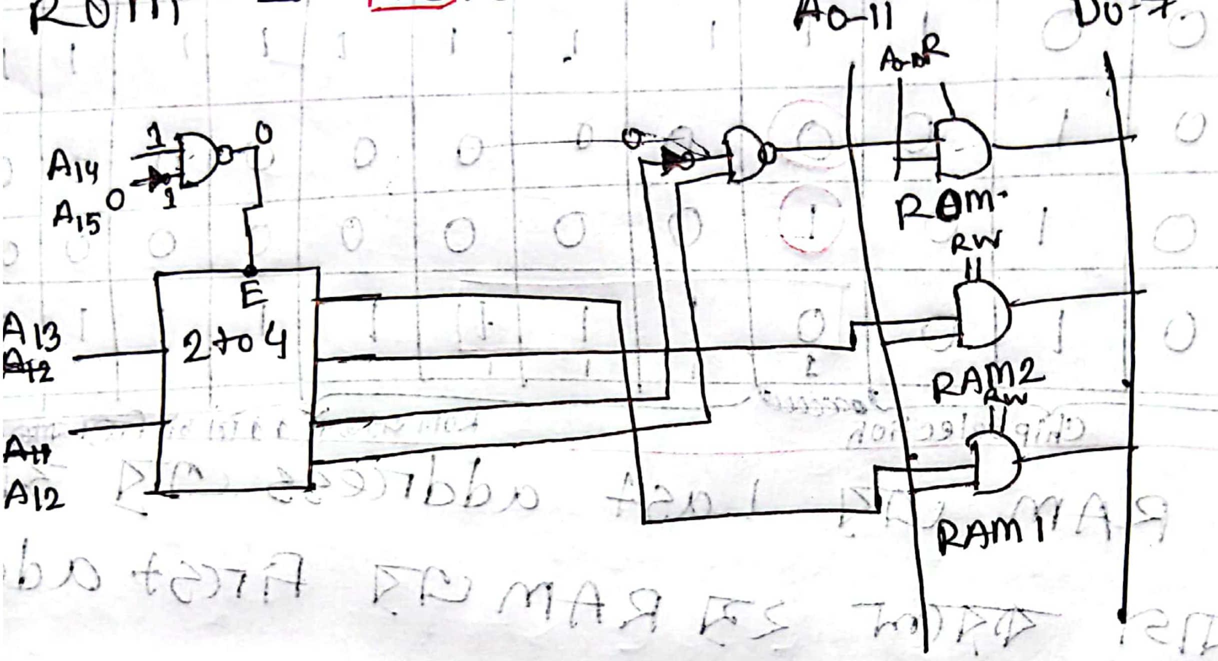
*ROM এর ক্ষেত্রে A_{0-10} পর্যন্ত 11 bit কাজ করবে। আগে থেকেই আমরা জানি A_{12-15} chip selection এর জন্য। তাহলে A_{11} হবে don't care condition এ থাকবে।

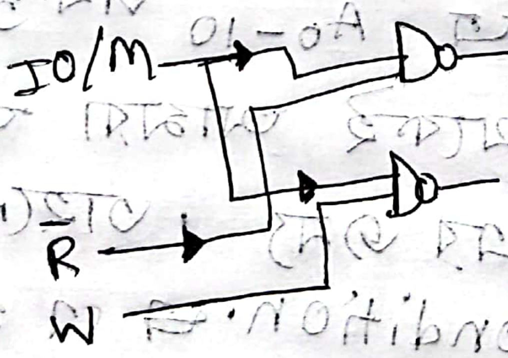
যদি A_{11} এ don't care condition এ 0 থাকে তাহলে starting address A000 এবং 1 হলে A800 chip selection:

RAM 1 — 1000

RAM 2 — 1001

ROM — 1010





figure

> Interface করা হয় latches এবং buffers দিয়ে. এটা I/O Interfacing বলে.

* I/O Interfacing:

There are three ways of transferring data between microcomputer and physical I/O device.

⇒ Programmed I/O: CPU নিজে কাজ করার মধ্যে instruction send করবে.

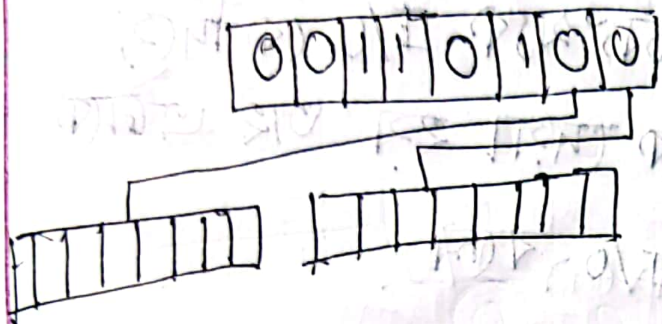
⇒ Interrupt driven I/O: External device CPU কে instruction send করবে to communicate

⇒ DMA: Direct Memory Access.

~~Programmed I/O~~ I/O ports are two types
serial

	Bit position	7	6	5	4	3	2	1	0
If: 0-R, 1-W	Data-direction register	0	0	1	1	0	1	0	0
	Data Register/Port Register	R	R	W	W	R	W	R	R
		↑	↑	↓	↓	↑	↑	↑	↑

Parallel



প্রত্যেক 8 bit এর সাথে একেকটা port connected

প্রত্যেক position একেকটা port কে control করে।

*Programmed I/O:

i) Isolated I/O - pin দিয়ে বোঝা যায় I/O use হচ্ছে

ii) Standard I/O

iii) Port

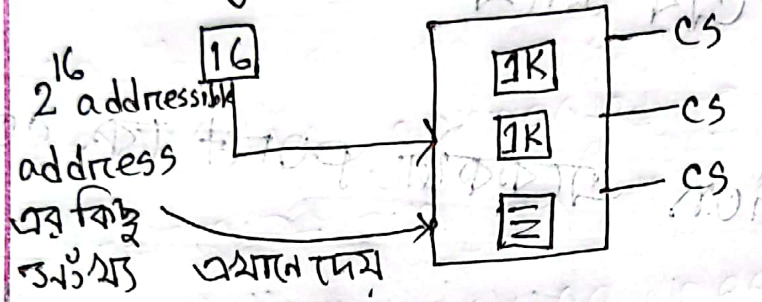
$IO/\overline{M}$ এই pin থাকলে standard

1 - I/O

0 - M

isolated এর সাথে specific register থাকতে হয় সেগুলো ছাড়া কাজ করে না।

* Memory Mapped I/O: total addressable address থেকে কিছু সংখ্যক I/O কে দিয়ে বাকিটা memory কে দেয়া হয় তাই এটাকে memory mapped I/O বলে।



* memory mapped I/O এ different register use করা হয়।

* Difference between standard I/O and memory mapped I/O:

Standard I/O	Memory Mapped I/O.
1. Memory and I/O have speparate address space.	1. Memory and I/O have same address space.
2. All address can be used by memory.	2. Due to addition of I/O addressable memory become less for memory.
3. More efficient due to separate bus.	3. Less efficient.

Standard I/O	Memory Mapped I/O
4. Larger in size	4. Smaller in size
5. It is complex.	5. It is simple
6. Use IO/$\overline{M}$ control pin	6. Does not use IO/$\overline{M}$ control pin.

MSB — 1 হলে I/O port select হবে
MSB — 0 " memory location accessed
হবে।

* Unconditional I/O: The microprocessor can send data to an external device at any time during unconditional I/O. The external device must always be ready for data transfer. Power consumption is high.

* Conditional I/O: The microprocessor outputs data to an external device via handshaking. Data transfer occurs by exchanging of control signals between

microprocessor and external device. Data transfer takes place when device is ready.

* Interrupt driven I/O

The external device is connected to a pin called Interrupt pin on the microprocessor chip. When device needs an I/O transfer with microprocessor or microcomputer, it activates interrupt pin.

Interrupt

- External → can be initiated through interrupt pin
  - Maskable → Enabled or disabled by executing instruction
  - Non-Maskable → has high priority, has power failure circuitry
- Internal → are acti by excep conditions l overflow
  - Trap → Like exception handling
  - Softwar → OS কে ca করা হলে

একটি interrupt এর জন্য processor এ একাধিক int pin থাকতে পারে। কাকে priority দিবে তা Depend করে –

→ Poll interrupt : depends on processor

→ Daisy chain interrupt : যাকে processor পাঠাবে এর পরের priority-র গুলোকে পাঠাবে।

* DMA :

Direct Memory Access is a technique that transfers data between a microcomputer memory and I/O device without involving microprocessor. DMA controller chip is used for data transfer operation. The main function are :

→ The I/O device request DMA operation via DMA request lines of the controller chip.

→ The controller chip activates the microprocessor HOLD pin, requesting CPU to release bus.

→ The processor sends HLDA (Hold acknowledges) back to DMA controller, indicating that the bus is disabled. The DMA controller places current value of its internal register on the system bus and sends DMA acknowledge to the peripheral device. DMA controller completes the DMA transfer and release buses.